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Hardware

GIGA Display Shield



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SquareLine Studio With the GIGA Display Shield

Learn how to use SquareLine Studio to create LVGL GUIs for the GIGA Display Shield

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The GIGA Display Shield with the GIGA R1 WiFi board can run LVGL which allows for the creation of advanced GUIs. To make the design of the GUI easier we can use the software SquareLine Studio. This will allow us to drag and drop different elements and then export it for usage on the display shield.

Hardware & Software Needed

[Arduino GIGA R1 WiFi](#)[Arduino GIGA Display Shield](#)[Arduino IDE](#)[SquareLine Studio](#)

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Hardware & Software Needed

[Downloading the Library and Board Package](#)[Using SquareLine Studio](#) +[Using SquareLine Studio Widgets in Arduino IDE](#) +[Rotating the SquareLine UI](#)[Next Step](#)

Downloading the Library and Board Package

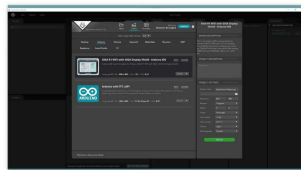
Make sure the latest GIGA Board

[Help](#)

Manager.... Here you need to look for the **Arduino Mbed OS Giga Boards** and install it, the **Arduino_H7_Video library** is included in the Board Package. Now you have to install the library needed. Go to **Tools > Manage libraries..**, search for **lvgl**(SquareLine studio is only compatible with LVGL version 8.3.11 or earlier, please keep the version that you install in mind here) and **Arduino_GigaDisplayTouch**, install both of these libraries.

Using SquareLine Studio

First download SquareLine Studio from the link above. Now in SquareLine Studio go to the "create" section and select the Arduino tab. Here you can select the GIGA Display Shield as a preset option so the project will be set up for the board and shield. Keep note of the LVGL version selected here on the right side.



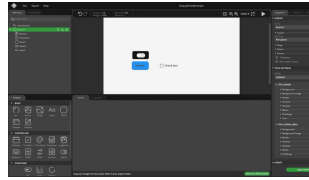
Project settings in
SquareLine Studio

Finally click on the "create" button.

Creating the GUI

Now it is time to create the GUI that will be on the display shield. Feel

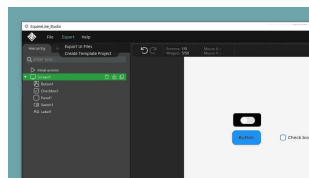
plenty of handy widgets that can be drag and dropped. On the right you can set more specific options for the widgets.



Project in SquareLine Studio

Exporting a Project From SquareLine Studio

Now that the project is ready it is time to export it. First click on the "Export" tab at the top, then select "Create Template Project" and select the destination for your exported files. When that is done click on the "Export UI Files" and wait for the process to finish.

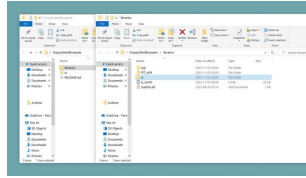


Export options in SquareLine Studio

Now all the files should be in the folder that you specified.

Running the SquareLine Studio Project on Arduino IDE

In the folder that you exported the files to, go into the "libraries" folder and then copy the "ui" folder.



Folder structure of the
exported files

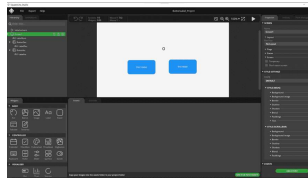
Place this folder in your **Libraries** folder found inside your **Arduino** folder. This is the same **Arduino** folder that contains your Arduino IDE sketches.

Now to run the SquareLine Studio project, use this sketch:

```
1  #include "Arduino_H7_Vid
2  #include "Arduino_GigaDi
3
4  #include "lvgl.h"
5  #include "ui.h"
6
7  /* Insert resolution WxH
8  Arduino_H7_Video
9  Arduino_GigaDisplayTouch
10
11 void setup() {
12     delay(3000);
13     Display.begin();
14     Touch.begin();
15
16     ui_init();
17 }
18
19 void loop() {
20
21     /* Feed LVGL engine */
22     lv_timer_handler();
23 }
```

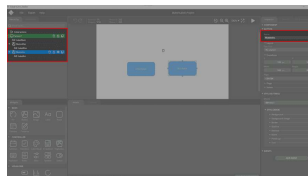
Using SquareLine Studio Widgets in Arduino IDE

The project from SquareLine Studio is exported as a library, let's take a look at how we can reference a specific element in the GUI. To demonstrate this we will first create a simple GUI with a increase button, a decrease button and a label. The label will show the number that will decrease or increase depending on what button is pressed. Our GUI will look like this:



The GUI we will be using
in this section

Pay attention to the names of the button and the label, which can be seen on the right side and the upper left side here:



Name sections
highlighted in SquareLine
Studio

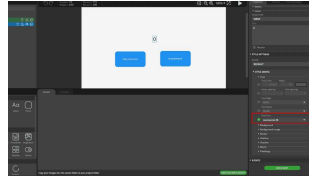
Name the widgets accordingly:

Increase Button: ui_ButtonInc

Decrease Button:
ui_ButtonDec

Number Label: ui_LabelNum

Also pay attention to the size of the font set for the counter label. We set the size to 26, this then needs to be enabled in the lv_conf.h file. This file can be found in the



Font size option in
SquareLine Studio

And to enable the font size find the **FONT USAGE** section. If you want to enable any other size simply change the **0** next to any of the font sizes into a **1**:



lv_conf.h font usage
section

Now export the project and put the library in the Arduino libraries folder, as shown in the previous section.

First declare the libraries and set up the screen, this will be the same as the sketch above.

```
1 #include "Arduino_H7_Vide
2 #include "Arduino_GigaDis
3
4 #include "lvgl.h"
5 #include "ui.h"
6
7 /* Insert resolution WxH
8 Arduino_H7_Video
9 Arduino_GigaDisplayTouch
```

Then it is as simple as using the names of the widgets in a LVGL function. For example, when a

```
lv_label_set_text_fmt(NAME_OF_LABEL, "Label");
```

needs a reference to a label object, we can enter the name of the label that we created in SquareLine Studio. Same goes for the button widgets, like this in the `setup()` function:

```
1 void setup() {
2     delay(3000);
3     Display.begin();
4     Touch.begin();
5
6     /* Initialize the ui
7     ui_init();
8
9     /* Add buttons event
10    lv_obj_add_event_cb(
11    lv_obj_add_event_cb(
12
13    /* Set initial value
14    label_val = 0;
15    lv_label_set_text_fm
16 }
```

The last important part of the sketch are the callback functions that will run when a button is pressed. In the segment above you can see how the callback functions are tied to the buttons. The function will check when the button is pressed and increase the count on the label:

```
1 static void ButtonInc_evt
2     lv_event_code_t code
3
4     if(code == LV_EVENT_C
5         label_val++;
6         lv_label_set_text
7     }
8 }
```

Full Sketch

```
1  #include "Arduino_H7_Vi
2  #include "Arduino_GigaL
3
4  #include "lvgl.h"
5  #include "ui.h"
6
7  /* Initialize the GIGA
8  Arduino_H7_Video Displa
9  Arduino_GigaDisplayTouc
10
11  int label_val;
12
13  static void ButtonInc_e
14      lv_event_code_t cod
15
16      if(code == LV_EVENT
17          label_val++;
18          lv_label_set_te
19      }
20  }
21
22  static void ButtonDec_e
23      lv_event_code_t cod
24
25      if(code == LV_EVENT
26          label_val--;
27          lv_label_set_te
28      }
```

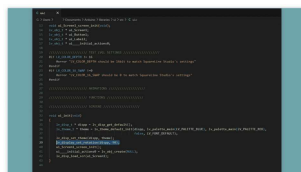
Rotating the SquareLine UI

If there is a need to rotate the UI created in SquareLine on the GIGA Display Shield, there is a single line that needs to be added to one of the exported library files. In the `/ui/src` folder, find the `ui.c` file and open it.

The code we need to add is:


```
1 lv_display_set_rotation(d
```

The `rotation` can be set to `0`, `90`, `180` or `270` depending on how much you want to rotate the UI.



Rotation code added to
ui.c file

Next Step

If you are curious about how LVGL works with the GIGA Display Shield, take a look at our [LVGL Guide](#).

LVGL can also be used in combination with the sensors on the display shield. Have a look at our [Image Orientation tutorial](#).

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